



OFFICIAL PROJECT SYNOPSIS & ARCHITECTURE SPECIFICATION

Full-Stack Multi-Tenant Freelance Marketplace & Talent Platform

A production-grade web application built to connect enterprise clients and independent professionals through role-segregated portals, a credit-metered bidding economy, real-time status management, and automated HTML email notifications.

React 19

Vite

Node.js (ESM)

Express.js 5

MongoDB NoSQL

Mongoose 9

Nodemailer HTML Engine

Bootstrap 5

Bcryptjs Security

PROJECT TITLE

Zentora Freelance Marketplace

DOMAIN

Gig Economy & Talent Acquisition

ARCHITECTURE

Client-Server RESTful Micro-SPA

TARGET ROLES

Client (Employer), Freelancer, Platform Admin

DATABASE INSTANCE

mongodb://localhost:27017/zentora

BACKEND PORT

RESTful API on Port 9000

1. Executive Summary & Problem Statement

Zentora is a full-stack digital marketplace created to solve critical operational bottlenecks in modern talent acquisition. Traditional gig platforms suffer from high middleman commissions, proposal spamming, delayed communications, and a lack of transparency between contract negotiation and execution.

Core Value Innovation

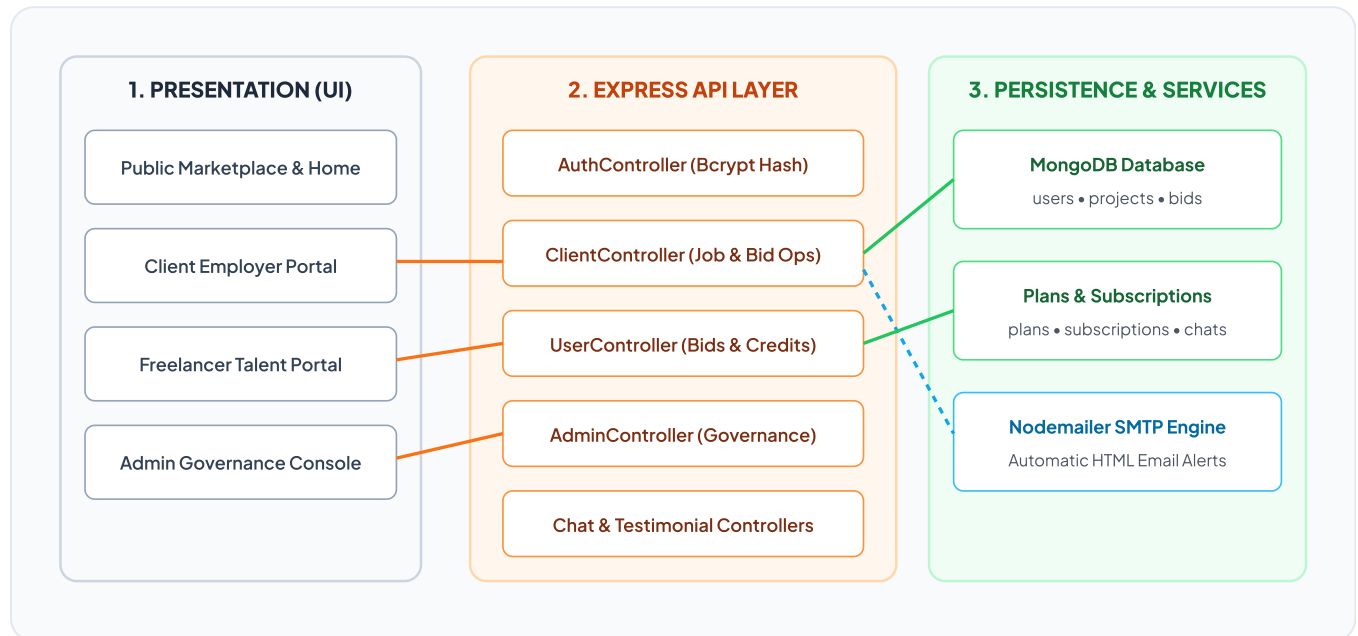
Zentora introduces a **Credit-Metered Proposal Economy** where freelancers must possess bidding credits to submit proposals, thereby deterring automated bot spam and encouraging high-intent, tailored bids. On proposal acceptance by a client, the backend automatically triggers a rich HTML email dispatch containing verified employer coordinates directly to the freelancer's Gmail inbox.

Key Functional Objectives

PILLAR	SYSTEM IMPLEMENTATION	TARGET USER ROLE
Role Segregation	Strict route protection with role verification preventing cross-portal privilege escalation.	All Roles (Admin / Client / User)
Project Lifecycle	End-to-end status pipeline from open marketplace bidding to contract assignment and closure.	Client & Freelancer
Credit Economy	Freelancers procure Master Plans; every submitted proposal deducts 1 credit from their live balance.	Freelancer
Direct Messaging	Dedicated project-scoped chat trail allowing client and hired freelancer to exchange requirements.	Client & Freelancer
Platform Governance	Instant user status toggle (active/blocked), soft/hard project moderation, and plan publishing.	Administrator

2. System Architecture & Data Flow

The system follows a decoupled Client-Server architecture. The frontend Single-Page Application (SPA) communicates asynchronously with Express 5 RESTful endpoints, while MongoDB persists documents via Mongoose schemas.



3. Complete Database Schema & Data Dictionary

The persistent layer runs on MongoDB via Mongoose 9 ODM schemas in `API/Module/module.js`.

3.1 Users Collection (users)

FIELD NAME	DATA TYPE	CONSTRAINTS / DEFAULTS	FUNCTIONAL DESCRIPTION
<code>_id</code>	ObjectId	Primary Key	Unique user document identifier
<code>name</code>	String	Required	Full personal name or company name
<code>email</code>	String	Required, Unique, Indexed	Primary login and notification coordinate
<code>password</code>	String	Required	Bcrypt salted hash with cost factor 10
<code>type</code>	String	'client' 'user' 'admin'	Role discriminator determining portal access
<code>credit</code>	Number	Default: 0	Available bidding balance for freelancers
<code>status</code>	String / Bool	Default: 'active'	Account governance ('active' or 'blocked')
<code>headline, bio, rate, skill</code>	String	Optional, Defaults: ""	Freelancer professional profile and rate card
<code>company, department</code>	String	Optional, Defaults: ""	Employer business organization metadata
<code>createdAt</code>	Date	Default: <code>Date.now</code>	Account registration timestamp

3.2 Projects Collection (projects)

FIELD NAME	DATA TYPE	CONSTRAINTS / DEFAULTS	FUNCTIONAL DESCRIPTION
<code>_id</code>	ObjectId	Primary Key	Unique project vacancy identifier
<code>clientId</code>	String (FK)	References <code>users._id</code>	Foreign reference to the client posting the job
<code>title</code>	String	Required	Project heading / contract title

FIELD NAME	DATA TYPE	CONSTRAINTS / DEFAULTS	FUNCTIONAL DESCRIPTION
desc / description	String	Required	Scope of work, deliverables, and technical requirements
budget	String	Required	Project budget range or fixed cost (e.g. ₹85,000)
duration / time	String	Optional	Estimated delivery timeframe (e.g. 1 Month)
status	Mixed	Default: false (Open)	Values: false (Open), 'closed', 'rejected by admin'
assignedTo	String (FK)	References users._id	Freelancer awarded the project contract
closedAt	Date	Optional	Timestamp when proposal was accepted and closed

3.3 Bids Collection (bids)

FIELD NAME	DATA TYPE	CONSTRAINTS / DEFAULTS	FUNCTIONAL DESCRIPTION
_id	ObjectId	Primary Key	Unique bid submission identifier
userId	String (FK)	References users._id	Freelancer tendering the proposal
projectId	String (FK)	References projects._id	Target project vacancy receiving the proposal
amount	String	Required	Proposed contract financial quote
message	String	Default: ""	Cover pitch, strategy, and deliverable commitments
status	String	Default: 'pending'	Lifecycle: 'pending' 'accepted' 'rejected'
createdAt	Date	Default: Date.now	Bid submission timestamp

3.4 Master Plans (plans) & Subscriptions (subscriptions)

COLLECTION	KEY ATTRIBUTES	DATA TYPES	FUNCTIONAL PURPOSE
plans	name, price, credits, tagline, features, popular, status	String, String, String, String, String, Boolean, Boolean	Admin-configured proposal credit bundles (e.g. Starter 10 Credits @ ₹99)
subscriptions	userId (FK), planId (FK), status, createdAt	String, String, Boolean, Date	Transaction audit ledger tracking user plan procurements

3.5 Direct Chat Messages (chatMessages)

FIELD NAME	DATA TYPE	CONSTRAINTS / DEFAULTS	FUNCTIONAL DESCRIPTION
projectId	String (FK)	Required	Project context linking client and freelancer
senderId	String (FK)	Required	Author user ID
senderName, senderRole	String	Defaults: 'User', 'user'	Display credentials and role badge

FIELD NAME	DATA TYPE	CONSTRAINTS / DEFAULTS	FUNCTIONAL DESCRIPTION
message	String	Required, Trimmed	Message body payload
createdAt	Date	Default: Date.now	Message delivery timestamp

4. Comprehensive RESTful API Reference

The Express 5 server exposes structured endpoints grouped by operational domain on port 9000.

4.1 Authentication & Profile Endpoints

METHOD	ENDPOINT	PAYLOAD / PARAMETERS	RESPONSE & DESCRIPTION
POST	/register	{ name, email, password, type }	Registers user with Bcrypt hash; returns created document without password.
POST	/login	{ email, password }	Verifies hash with <code>bcrypt.compare</code> ; returns user object for session storage.
GET	/get-profile	?userId={id}	Fetches full user profile details excluding sensitive password hash.
PUT	/update-profile	{ userId, name, headline, rate, skill... }	Updates profile metadata or rotates account password.

4.2 Client Employer Endpoints

METHOD	ENDPOINT	PAYLOAD / PARAMETERS	RESPONSE & DESCRIPTION
GET	/client-stats	?clientId={id}	Returns <code>totalProjects</code> , <code>totalBids</code> , and <code>acceptedBids</code> .
POST	/client-post-project	{ clientId, title, desc, budget, duration }	Publishes a new project into the open marketplace.
GET	/client-project-list	?clientId={id}	Fetches all projects posted by the specific client.
GET	/client-biding-list	?projectId={id}	Fetches all received proposals along with full bidder profile details.
PUT	/client-biding-action	{ projectId, userId, status }	Accepts proposal, auto-closes project, rejects rivals, and sends HTML acceptance email.
DELETE	/client-delete-project/:id	Route param <code>:id</code>	Cascading delete for project and associated proposals.

4.3 Freelancer Talent Endpoints

METHOD	ENDPOINT	PAYLOAD / PARAMETERS	RESPONSE & DESCRIPTION
GET	/user-stats	?userId={id}	Returns credits balance, valid bids count, accepted projects, and total earnings.
GET	/user-projects-list	None	Fetches open marketplace contracts excluding projects rejected by admin.
POST	/user-create-bids	{ userId, projectId, amount, message }	Validates credit > 0, deducts 1 credit, and logs proposal in bids.
GET	/user-get-bids	?userId={id}	Retrieves all proposals placed by the user with real-time status and client names.
POST	/user-purchase-plan	{ userId, planId }	Credits account balance with plan quota and creates audit subscription record.

4.4 Admin Governance Endpoints

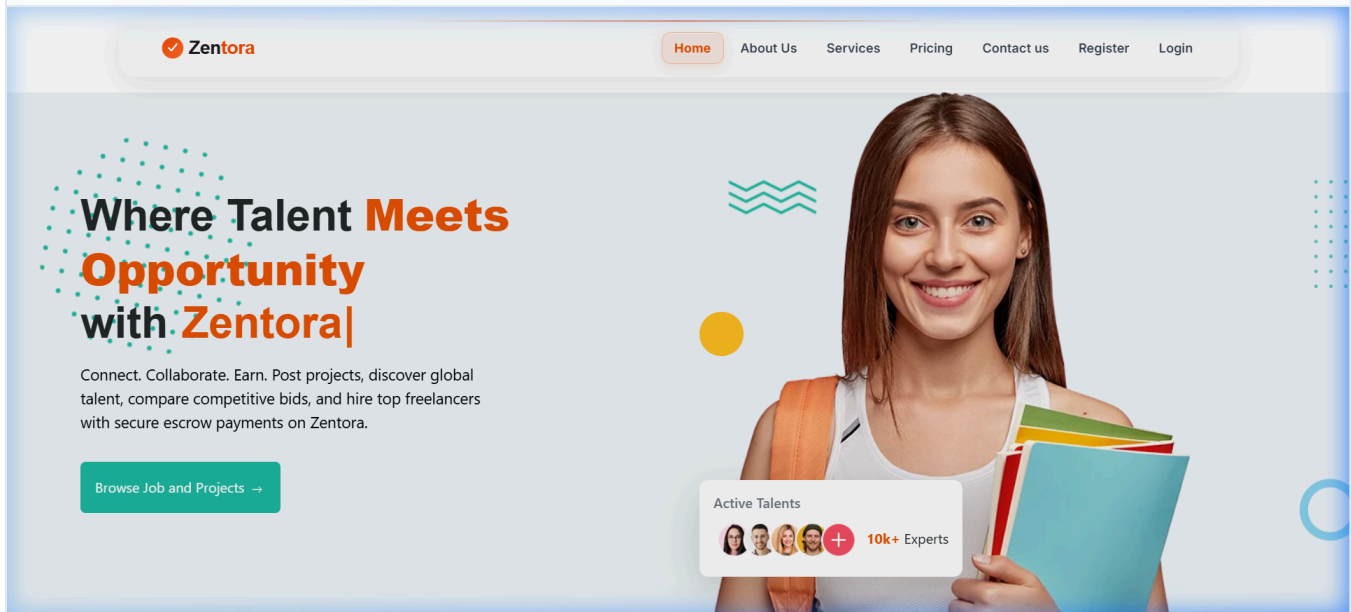
METHOD	ENDPOINT	PAYLOAD / PARAMETERS	RESPONSE & DESCRIPTION
GET	/admin-stats	None	Aggregate platform counts: total users, clients, and projects.
GET	/admin-users-list	None	Lists all registered freelancer profiles without sensitive passwords.
PUT	/admin-toggle-user-status/:id	Route param :id	Toggles account status between 'active' and 'blocked'.
PUT	/admin-reject-project/:id	Route param :id	Sets project to 'rejected by admin', shielding marketplace from spam.
POST	/admin-create-plans	{ name, price, credits, tagline... }	Creates new master pricing tier for freelancer proposal credit acquisition.

5. Live Application Output & Interface Gallery

Screenshots captured from the running Zentora application on <http://localhost:5173>.

1. Public Marketplace Landing Page & Hero Section

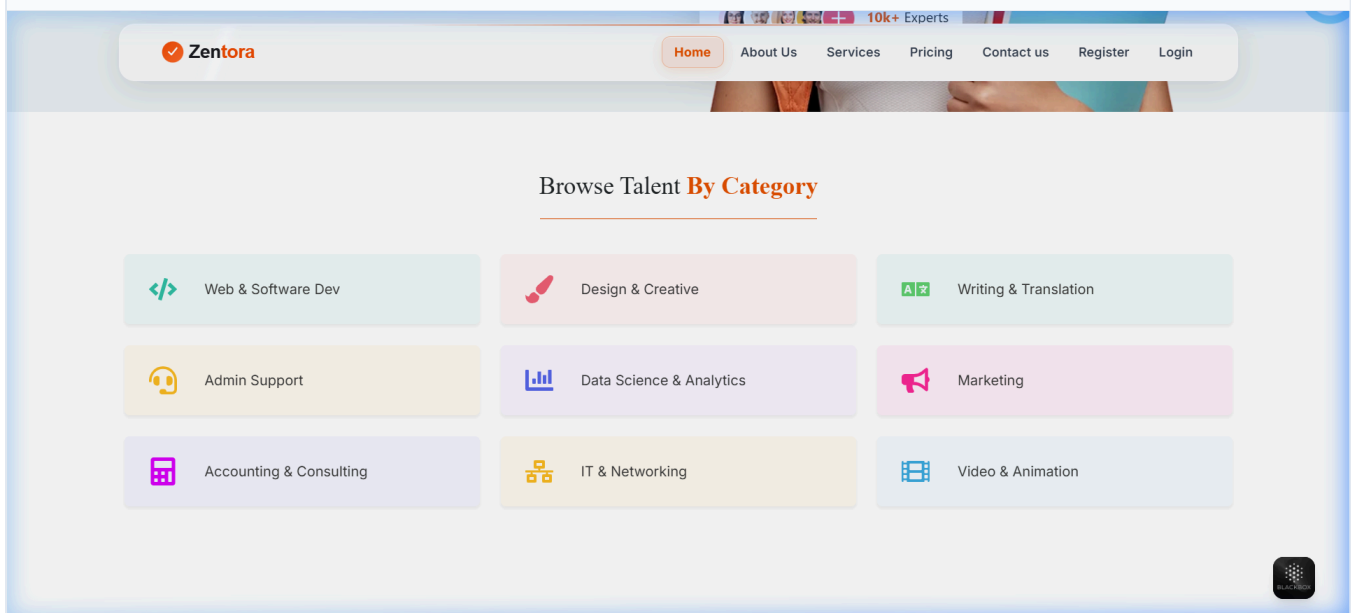
URL: /



The landing view presents brand identity, full-service navigation header, responsive CTA buttons, and high-visibility community counters showcasing 10k+ vetted experts.

2. Browse Talent by Categories & Platform Trust Pillars

URL: /#categories



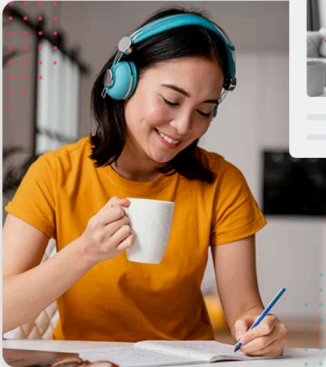
Nine professional talent sectors (Web & Software, Design, Writing, Data Science, Marketing, etc.) accompanied by core trust pillars: Expert Freelancers, Safe Escrow, and 24x7 Support.

3. Trending Contracts & Marketplace Open Project Feed

URL: /#projects

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Trusted By **Teams**



Work Without Borders With Zentora

Zentora is a secure, escrow-backed platform where businesses and freelancers connect across the globe. Hire. Work. Grow. — get premium development, design, content, and marketing delivered on time, within budget.

- Expert Freelancers
- Safe Escrow Payments
- 24X7 Priority Support

 **150+**
Top Countries

Live job opportunities showing employer parameters including contract heading, duration tags, budget quotations, and instant proposal submission triggers.

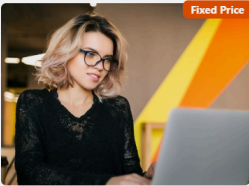
4. Platform Growth Metrics & Verified Client Reviews

URL: /#testimonials

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 Top Countries

Trending **Contracts**



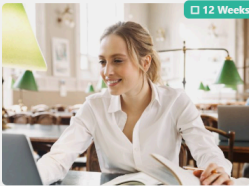
Fixed Price

Web Dev

E-commerce Website Dev (React & Node)

★★★★★ (5.0/ 3 Ratings)

₹85,000



12 Weeks

Expert

Grow Personal Financial Security Thinking &...

★★★★★ (5.0/ 2 Ratings)

₹45,000



20 Hours

All Levels

The Complete Guide to Build RESTful API...

★★★★☆ (4.0/ 2 Ratings)

Free



25 Hours

All Levels

Competitive Strategy Law for Management...

★★★★☆ (3.0/ 2 Ratings)

₹32,000

Production KPI counters (32k+ Freelancers, 8,768+ Contracts, 100% Satisfaction) paired with dynamic client and service provider testimonials.

5. Services Catalog

/services

Zentora

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OUR MARKETPLACE SERVICES

Comprehensive Freelance Solutions

Discover top-rated talent across key tech, design, marketing, and creative disciplines ready for contract hiring.

Web & Software Development

Custom web apps, React/Node.js solutions, e-commerce stores, API integrations, and full-stack software development.

UI/UX & Product Design

User interface design, Figma wireframing, mobile app designs, branding systems, and modern web graphics.

Mobile App Development

Cross-platform iOS and Android apps using React Native, Flutter, and native mobile technologies.

Digital Marketing & SEO

Search engine optimization, Google Ads campaigns, social media growth strategies, and conversion analytics.

Content Writing & Translation

Data Science & Analytics

Branding & Social Media

Cybersecurity & DevOps

Specialized service categories for web development, UX/UI, digital marketing, and DevOps.

6. Credit Pricing Packages

/pricing

Zentora

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FOR FREELANCER CREDIT TIER / PLAN

✓ Priority bid visibility

○ Standard analytics

Get Plan

10 credits included | 1 credit per bid

Zentora

Where talent meets opportunity. The premier freelance marketplace for global collaboration, verified expertise, and secure income contracts.

Location: Lucknow, India

Support Phone: +91 7266027178

Email: support@zentora.com

Our Platform

- About Us
- Browse Projects
- Post a Project
- Freelancer Credit Plans
- Marketplace Services

Quick Links

- Home
- Services
- Pricing
- Contact Us
- Sign In
- Create Account

Newsletter

Subscribe to our newsletter for featured freelance projects, client hiring tips, and platform updates.

Subscribe

Freelancer credit tiers (e.g. Starter Plan: 10 Credits @ ₹99) enforcing proposal intent.

7. User Sign-In Portal

/login

Zentora

HomeAbout UsServicesPricingContact usRegisterLogin

WELCOME BACK

Login

Don't have an account? [Create one free](#)

Email Address

you@example.com

Password

Enter password

Remember me

Sign In

Secure authentication with real-time validation and automatic role-based redirect.

8. Registration & Role Choice

/register

Zentora

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GET STARTED

Create Account

Already have an account? [Sign in here](#)

Register As

Client — I want to hire

Full Name / Company

Your name or company

Email Address

you@example.com

Password

Min. 6 characters

Confirm Password

Re-enter password

Create Account

Admin accounts are created by the platform team only — not available via registration.

Account creation interface supporting role segregation between Client and Freelancer.

6. Technical Roadmap & Conclusion

Planned Enhancements

ENHANCEMENT	TECHNICAL ROADMAP STRATEGY	TARGET MILESTONE
Payment Escrow	Integration with Razorpay / Stripe Webhooks for automated INR/USD milestone deposits and fund releases.	v2.1 Release
WebSocket Chat	Migrate HTTP chat polling to bidirectional Socket.io with live typing indicators and presence detection.	v2.2 Release
AI Proposal Assistant	Embedding LLM capability to help freelancers draft tailored proposal pitches based on job specifications.	v2.3 Release
Mobile App	Deploying React Native cross-platform mobile apps for iOS and Android consuming existing REST APIs.	v3.0 Release

Conclusion

The **Zentora Freelance Marketplace** provides an end-to-end, dependable, and visually captivating talent ecosystem. By uniting strict role-based routing, spam-deterrent credit bidding, automated HTML email dispatch via Nodemailer, project-scoped collaboration hubs, and administrative governance, Zentora establishes a reliable foundation for modern online freelancing.

End of Document • Zentora Project Synopsis

Compiled using React 19 • Vite • Node.js • Express 5 • MongoDB • Nodemailer